

Fundamentals of Network Management

- **Network management** means checking, monitoring, and taking care of all parts of the network.
- What you manage depends on the **size and importance** of the network.
- There are **different areas** in network management, but they all aim for the same results:
 - Make the network work better and faster.
 - Avoid **downtime** and prevent money or data loss.
 - Predict problems before they occur

! Baseline Measurements

- A **baseline** shows the **current condition of the network**.
→ It helps us understand how the network normally works.
- **Examples of baseline measurements** include:
 - How much of the **network backbone** is being used.
 - How many **users log in** daily or hourly.
 - How many **protocols** are active on the network.
 - **Error reports** like:
 - **Runts** (too small), **jabbers** (too long), or **giants** (oversized frames).
 - **Frequency** of application use
 - **Bandwidth** usage

! Baseline Measurements

- **Compare future and past performance**
→ Focus on important parts of the network and how users work
→ More data provides more accuracy.
- **Forecasting network traffic patterns**
→ It's hard to guess user behavior or future needs because of habits and new technologies
- **Gathering baseline data**
→ Use software tools (free ex: Freeware or paid) to collect and monitor network information.

Monitoring and Managing Network Traffic

- Continuously check the network's condition to make sure everything works well.
- **Performance management** means watching how links and devices handle the traffic and demand.
- **Fault management** means finding and reporting problems with devices, connections, or components.

! SNMP (Simple Network Management Protocol) Logs

- **Enterprise-wide network management systems**
 - Help with managing faults and performance management.
 - All use similar architecture
- Central collection point is called the **Network Management System (NMS)**
 - **Network management agent**
 - A software that collects data about how the device works
 - Sends this data to the NMS
 - **Managed device**
 - Any device in the network that the NMS watches
- **Device management**
 - Different parts of a device can be managed (like CPU, memory, hard disk, NIC, and even non-physical parts).
- **MIB (Management Information Base)**
 - A database that defines and stores information about managed devices.
- **SNMP (Simple Network Management Protocol)**
 - A protocol used to share information about managed devices.
 - It is part of the TCP/IP protocols.
 - **SNMPv3** is the most secure version.
 - **SNMPv2** is still widely used.
 - **SNMPv1** is the original version, rarely used today.

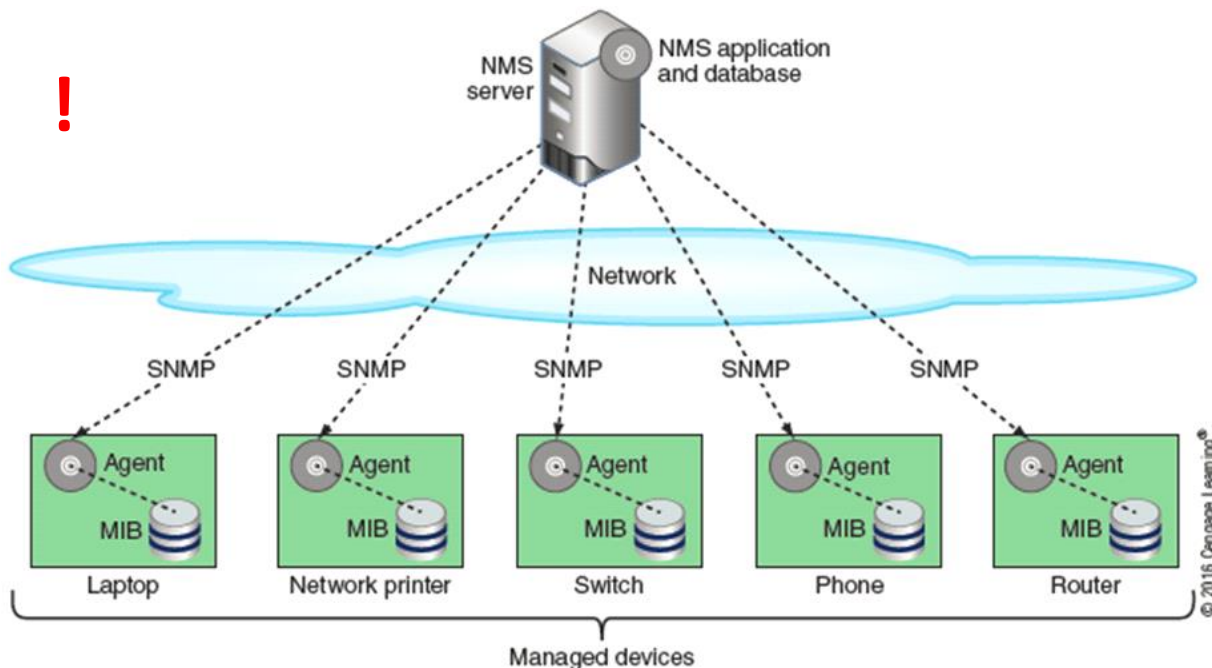


Figure 9-2 Network management architecture

This diagram shows the **network management architecture** using **SNMP**.

- This server **monitors and manages** the devices in the network.

How it works:

- Each device (like a **laptop, printer, switch, phone, router**) is a **managed device**.
- Every device has:
 - An **Agent**: A small software program that collects data.
 - A **MIB (Management Information Base)**: A database that stores information about the device (e.g. CPU, memory, etc.).
- The **Agent** sends information from the MIB to the NMS server using **SNMP**.

SNMP Logs

- There are different ways to see and analyze the data collected from network devices.
- **Network Management Applications:**
 - They are **flexible**,
 - But can be **hard to configure and fine-tune**,
 - It's important to **choose the correct type and amount of data** to collect, so the system works well.

! System and Event Logs

- **Log:** Stores information about events or issues found by the operating system.
- **Event log:** A special log in Windows that saves data about what happens on the computer.
- **Event Viewer application:** A Windows tool used to open and read these logs.
- **Syslog function:**
It is a standard system used in Linux and UNIX to **create, save, and manage messages about events**. These messages are stored in the system log.
- **Syslog roles:**
 - **Generator:** A computer that **sends event messages** using a syslog-supported tool.
 - **Collector:** A computer that **receives and stores** these **event messages** from generators.
- Logs are used to **track system history** and help with **fault management**.
- Collected log data does not point to problem, so **logs must be monitored carefully**.
- Most **UNIX and Linux systems** offer a **GUI tool** to view and filter logs.
- **Log file locations:**
 - New Linux: `/var/log/messages`
 - Older UNIX: `/var/log/syslog`
 - Solaris UNIX: `/var/adm/messages`

! Traffic Analysis

- **Network monitor:**
A tool that watches all network traffic all the time.
- **Interface monitor:**
A tool that watches traffic at a specific connection point (like between a server and the network).
- **Difference:**
 - Network monitor checks traffic for a single device.
 - Interface monitor checks traffic patterns across the network.

All network monitoring tools can perform following functions;

- Set the NIC in **promiscuous mode** to capture all traffic
- **Monitor traffic** continuously on a segment
- **Capture** data being transmitted
- **Capture frames** sent from or to a specific node
- **Simulate traffic** by sending custom data
- **Create statistics** about network activity

Some network monitoring tools can also do the following;



- Find all devices on a network segment
- Set a baseline for performance and usage
- Track resource usage and show it as charts or graphs
- Save traffic data and create reports
- Trigger alerts when specific conditions happen
- Detect unusual activity, like heavy users or listeners



Monitoring tools can detect problems like:

- Runts, giants, jabber, and ghost frames
- Packet loss and dropped packets
- Interface resets

Alerts about these issues can be sent by email or text (SMS – Short Message Service).

Traffic Management

Traffic shaping (packet shaping)

- Traffic shaping is used to control how data moves through a network.
- It adjusts packets, data flow, and connection features to manage the type and amount of traffic.

Goals

- Ensures fast delivery of the most important traffic
- Provides best performance for all users

Techniques

- Slow down less important traffic
- Give higher priority to more important traffic
- **Traffic policing**
 - Control traffic flow in and out during a set time
 - Limit short-term speed for a connection
- **Not without controversy**
 - Comcast blocked some types of traffic unfairly
- **Traffic prioritization**
 - Give better treatment to more important traffic
- **Based on characteristics**
 - Protocol
 - IP address
 - User group
 - DiffServ flag in IP packets
 - VLAN tag in Data Link frame
 - Service or application type

Caching

- ! • **Caching**
 - Saves frequently used files locally.
 - Makes file access faster.
- **Web Caching**
 - Most popular type of caching; it can be customized.
 - Web pages are saved locally:
 - Either on the host or within the network.
 - Then sent to users when requested.
- **ISP Cache Engine**
 - A special network device used for storing and delivering popular files.
 - Reduces internet traffic and saves money.

! Unified Communications Technologies

unified = birleşik

- **PSTN (Public Switched Telephone Network)**
 - Uses a circuit-switched model to transmit phone calls and faxes.
- **Switching**
 - Decides how connections are made between network nodes.
- **Circuit-switched networks**
 - A dedicated connection is set up between two nodes before transmitting data.
- **Packet-switched networks**
 - Data is divided into packets before being transported.
- **Main advantage of packet switching**
 - It saves bandwidth by not keeping the connection open until message reaches its destination.
- **Ethernet and Internet**
 - These are the most common examples of packet-switched networks.
- **With unified communications, users can:**
 - Use the web
 - Send and receive faxes, emails, voicemails, instant messages, or phone calls
 - Join video conference calls

VoIP Applications and Interfaces

- **IP telephony (VoIP)**
 - Uses any network to send voice signals via TCP/IP.
 - Can be public or private.
 - Works on any packet-switched network.
- **Types of data connections that support VoIP**
 - Includes T-carriers, ISDN, DSL, broadband cable, satellite, WiFi, WiMAX, HSPA+, LTE, and cellular networks.

VoIP Applications and Interfaces

- **Why use VoIP?**
 - Reduces the cost of voice calls
 - Offers new and improved features
 - Centralizes voice and data network management
- **Voice and Data Configurations**
 - Analog phones (send/receive analog signals) must connect to one of the following:
 - **ATA:** Analog Telephone Adapter
 - **Switch/router/gateway:** Converts analog signals to packets
 - **Digital PBX (IP-PBX) or Analog PBX**
- **Voice and Data Configurations**
 - **IP telephones:**
 - Send and receive only digital signals.
 - **Softphones:**
 - Software on a computer that works like an IP phone.

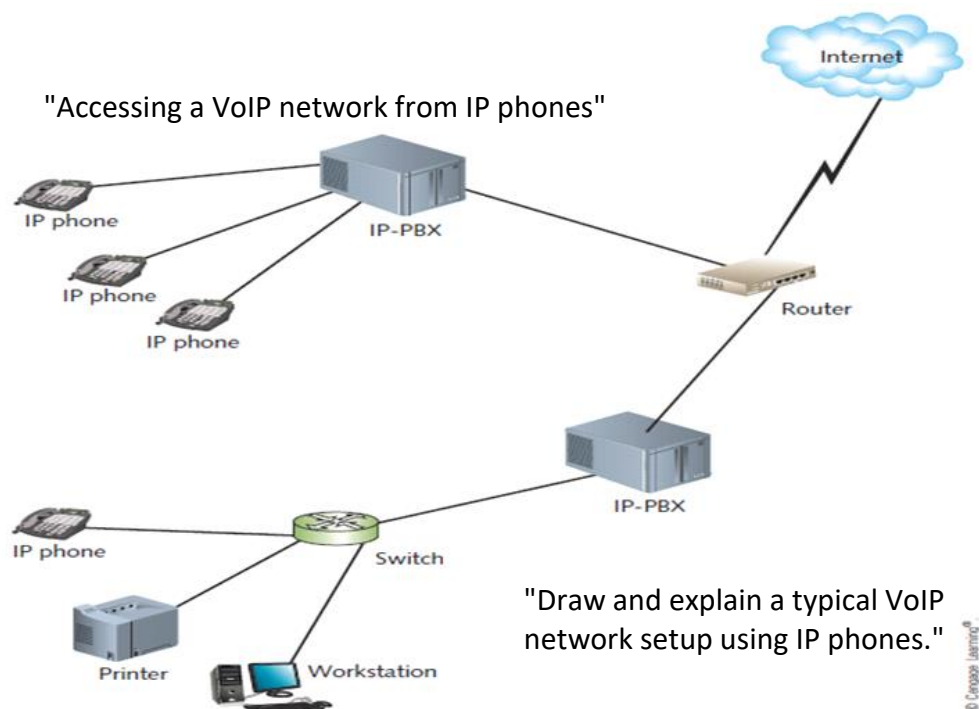


Figure 9-8 Accessing a VoIP network from IP phones

Video over IP Applications and Interfaces

- **Videoconferencing (VTC)**
 - Lets many users join and interact in real-time video meetings.
- **Cisco Systems Prediction**
 - In 2018, more than 79% of internet traffic was expected to be video.
- **Reasons for growth**
 - Huge amount of video content online
 - More devices are connected to the Internet
 - Bandwidth and equipment are becoming cheaper

Summary – Key Points

- **Baseline measurements** help compare past and future network performance to detect changes.
- **Network managers** must follow not only internal policies but also state and federal regulations.
- **Syslog protocol** is used to create, store, and manage messages about system events.
- **Using interface and event logs effectively** helps detect and avoid network issues early.